

P R O J E C T R E P O R T

Simulation of a Buck Converter

Using MATLAB/Simulink

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Abstract

DC–DC converters are widely used in power electronics to obtain a desired DC voltage level from a fixed DC source. Among various DC–DC converters, the **Buck Converter** is one of the most important because it efficiently reduces the input voltage to a lower output voltage. This project presents the design and simulation of a Buck Converter using MATLAB/Simulink. The converter is analysed under **Continuous Conduction Mode (CCM)**, and the effects of duty cycle on output voltage are investigated. Simulation results verify the theoretical operation of the converter while reflecting realistic semiconductor losses inherent in the simulation environment.

1. Introduction

Power electronic converters are essential components in modern electrical systems. They are used in renewable energy systems, battery-powered devices, electric vehicles, communication equipment, and industrial control systems.

A **Buck Converter** is a switching power supply that converts a higher DC voltage into a lower DC voltage while maintaining high efficiency. Unlike linear regulators, Buck Converters use semiconductor switches and energy-storage elements, resulting in significantly lower power losses.

The objective of this project is to model and simulate a Buck Converter using MATLAB/Simulink and to compare the theoretical and simulation results.

2. Objectives

- Understand the operating principle of a Buck Converter.
- Design and implement a Buck Converter model in MATLAB/Simulink.
- Study the effect of duty cycle on output voltage.
- Analyse output voltage and current waveforms.
- Compare simulation results with theoretical calculations.

3. Theoretical Background

Circuit Components

- DC Input Voltage Source
- MOSFET Switch (controlled by PWM gate signal)
- Freewheeling Diode
- Energy-Storage Inductor (L)
- Filter Capacitor (C)
- Load Resistance (R)

3.1 Operating Principle

When the MOSFET switch is **ON**, the DC source supplies energy to the load and the inductor stores magnetic energy. When the switch is **OFF**, the inductor releases its stored energy through the freewheeling diode, maintaining continuous current flow to the load.

3.2 Governing Equation

The average output voltage for an ideal Buck Converter is related to the input voltage by the duty cycle:

$$V_{out} = D \times V_{in}$$

where: V_{out} = Output Voltage, V_{in} = Input Voltage, D = Duty Cycle ($0 \leq D \leq 1$)

For an ideal converter, conversion efficiency is assumed to be 100 %. In practice, semiconductor conduction and switching losses reduce this value.

4. Circuit Parameters

The following parameters were selected for simulation:

Parameter	Symbol	Value
Input Voltage	V_{in}	24 V
Switching Frequency	f_s	20 kHz
Duty Cycle	D	0.5 (50 %)
Inductor	L	1 mH
Capacitor	C	220 μ F
Load Resistance	R	10 Ω
Simulation Time	t	0.1 s

5. Design Calculations

5.1 Output Voltage

Applying the fundamental Buck Converter relation:

$$\begin{aligned}V_{out} &= D \times V_{in} = 0.5 \times 24 \text{ V} \\ \therefore V_{out} &= 12 \text{ V}\end{aligned}$$

5.2 Output Current

By Ohm's Law:

$$\begin{aligned}I_{out} &= V_{out} / R = 12 \text{ V} / 10 \Omega \\ \therefore I_{out} &= 1.2 \text{ A}\end{aligned}$$

5.3 Output Power

$$\begin{aligned}P_{out} &= V_{out} \times I_{out} = 12 \text{ V} \times 1.2 \text{ A} \\ \therefore P_{out} &= 14.4 \text{ W}\end{aligned}$$

6. Continuous Conduction Mode (CCM) Verification

Continuous Conduction Mode is confirmed when the average output current exceeds half the peak-to-peak inductor ripple current:

6.1 Switching Period

$$T = 1 / f_s = 1 / 20\,000 \text{ Hz} = 50 \times 10^{-6} \text{ s} = 50 \mu\text{s}$$

6.2 Inductor Ripple Current

$$\begin{aligned}\Delta I_L &= [(V_{in} - V_{out}) \times D \times T] / L \\ \Delta I_L &= [(24 - 12) \text{ V} \times 0.5 \times 50 \times 10^{-6} \text{ s}] / (1 \times 10^{-3} \text{ H})\end{aligned}$$

$$\Delta I_L = 0.3 \text{ A}$$

6.3 CCM Condition

$$I_{out} > \Delta I_L / 2$$

$$1.154 \text{ A} > 0.15 \text{ A} \quad \checkmark \text{ CCM Confirmed}$$

∴ The converter operates in Continuous Conduction Mode, confirming stable inductor current throughout each switching cycle.

7. MATLAB/Simulink Model

The Buck Converter was implemented in MATLAB/Simulink using the **Simscape Power Systems** library. Key blocks used:

- DC Voltage Source (24 V)
- MOSFET Switch with PWM gate drive
- Pulse Generator — 20 kHz, 50 % duty cycle
- Freewheeling Diode
- Inductor (1 mH) and Capacitor (220 μ F)
- Load Resistance (10 Ω)
- Voltage and Current Measurement blocks
- Scope blocks for waveform observation

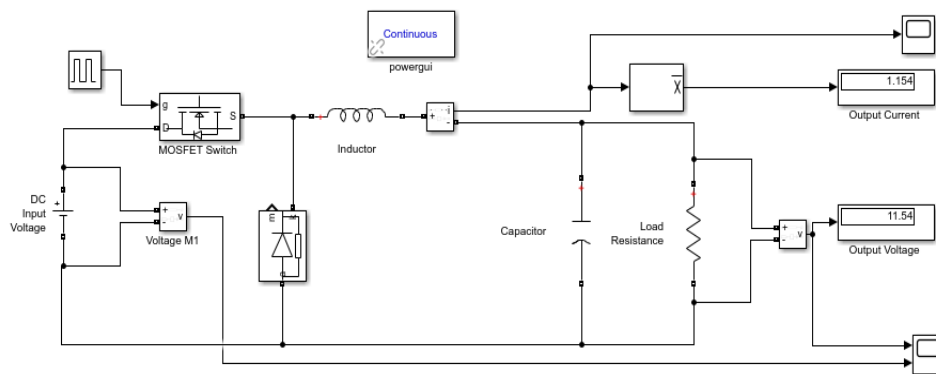


Figure 1. Complete MATLAB/Simulink Buck Converter Model.

8. Simulation Results

8.1 Output Voltage Waveform

During startup, the output voltage exhibits transient oscillations caused by the charging of the LC filter. The voltage reaches a peak of approximately **19.5 V** before gradually settling. After the transient period, the output voltage stabilises at approximately **11.54 V**, consistent with the theoretical prediction and the minor losses attributable to semiconductor devices.

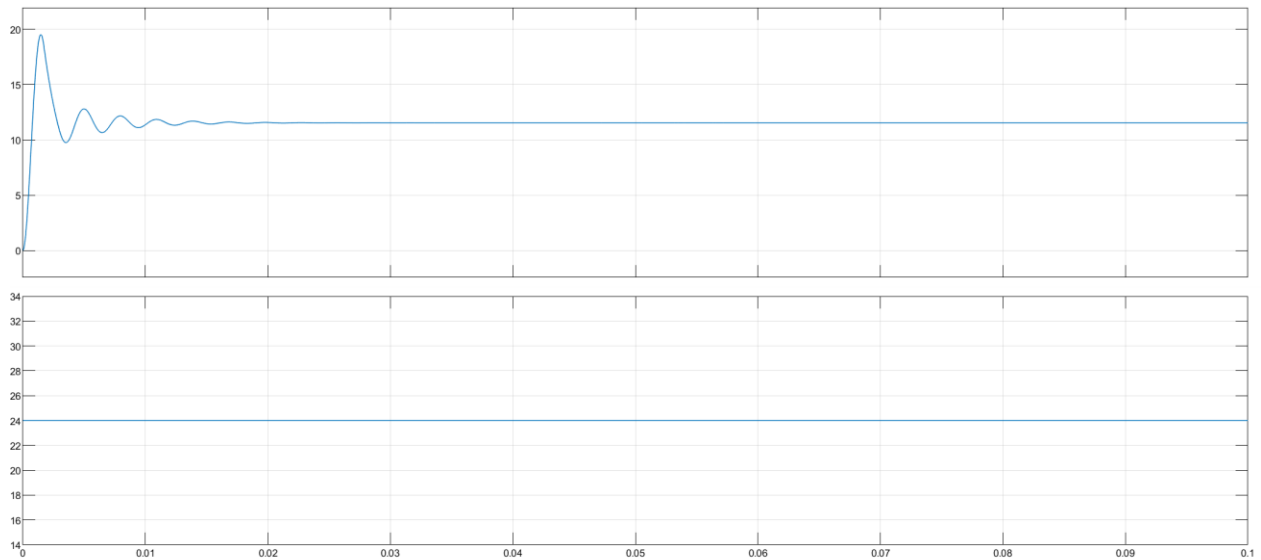


Figure 2. Output Voltage Waveform — Startup Transient and Steady-State Response.

8.2 Output Current Waveform

The output current initially experiences a transient surge of approximately **5.7 A** caused by capacitor charging and inductor energy build-up. The oscillations gradually decay and the current settles to a steady-state value of approximately **1.154 A**. The waveform confirms stable converter operation in CCM throughout the steady-state region.

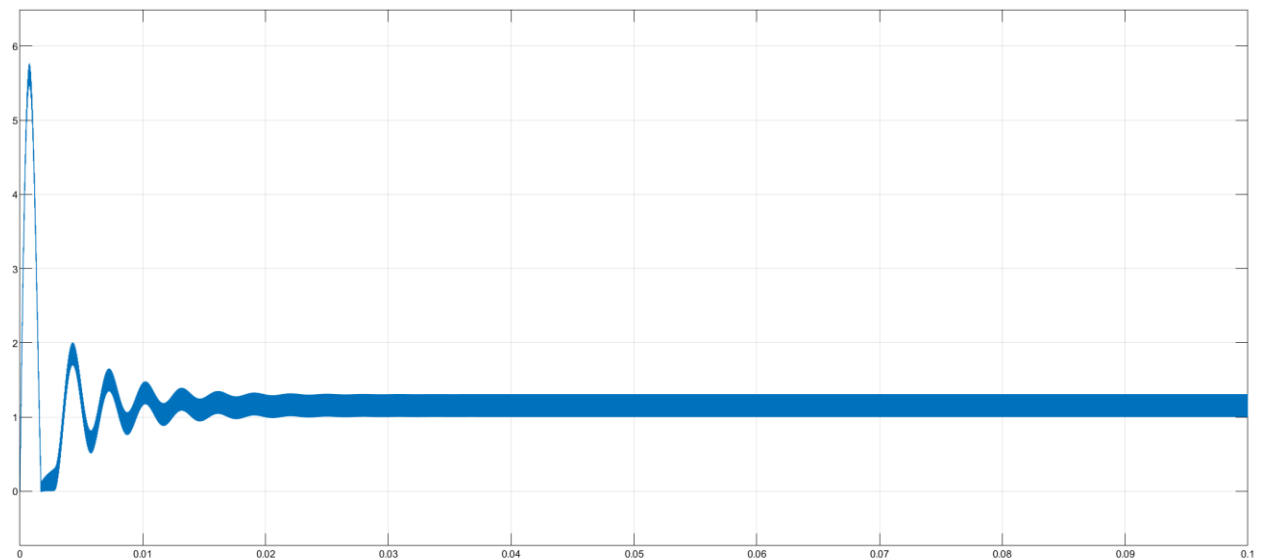


Figure 3. Output Current Waveform — Startup Transient and Steady-State Operation.

9. Results and Discussion

9.1 Comparison Table

Table 1 compares the theoretical calculations with the simulation results.

Parameter	Theoretical Value	Simulation Value	Error (%)
Input Voltage (Vin)	24 V	24 V	0.00%
Duty Cycle (D)	50%	50%	0.00%
Output Voltage (Vout)	12.00 V	11.54 V	3.83%
Output Current (Iout)	1.200 A	1.154 A	3.83%

Table 1. Comparison of Theoretical and Simulation Results.

9.2 Discussion

The simulation results show excellent agreement with the theoretical calculations. The output voltage deviates by only **3.83 %** from the ideal value. This deviation is expected because practical simulations include the diode forward-voltage drop and the MOSFET on-state resistance, both of which introduce conduction losses absent from the ideal model.

9.3 Efficiency Estimation

Using the simulation results, the actual output power and converter efficiency are calculated as follows:

$$P_{\text{out(sim)}} = V_{\text{out}} \times I_{\text{out}} = 11.54 \text{ V} \times 1.154 \text{ A} = 13.32 \text{ W}$$

$$P_{\text{out(ideal)}} = 14.4 \text{ W}$$

$$\eta = P_{\text{out(sim)}} / P_{\text{out(ideal)}} \times 100$$

$$\eta = 13.32 / 14.4 \times 100$$

$$\therefore \eta \approx 92.4 \%$$

The converter achieved an estimated efficiency of approximately **92.4 %**, demonstrating high-performance conversion and validating the Buck Converter topology for practical applications.

10. Advantages of Buck Converter

- High efficiency compared to linear regulators.
- Low conduction and switching losses.
- Compact form factor.
- Fast dynamic response.
- Suitable for battery-powered and portable systems.

11. Applications

- Electric Vehicles (EV traction and auxiliary systems)
- Solar Power Systems (MPPT controllers)
- Battery Chargers
- Industrial Control Systems
- Embedded and Microcontroller Systems

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- Communication and Networking Equipment

12. Conclusion

In this project, a Buck Converter was designed and successfully simulated using MATLAB/Simulink. The converter reduced the input voltage from **24 V** to approximately **11.54 V** while delivering a steady-state output current of **1.154 A**

The simulation results closely matched the theoretical calculations with an error of less than **4 %**. The minor deviation was attributed to realistic semiconductor losses — namely the diode forward-voltage drop and the MOSFET conduction resistance. Continuous Conduction Mode (CCM) operation was verified through inductor ripple current analysis, and the converter achieved an estimated efficiency of **92.4 %**.

This study successfully demonstrates the operating principles, performance characteristics, and practical applications of Buck Converters in modern power electronic systems.

References

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